

ABYSS

3D Psychological Horror Coding Escape Room & Adaptive CS Learning Platform

"The building has been empty for nine years. You came down here to steal a hard drive. The door locked behind you. Somewhere in the dark, a machine is still running — and it has been waiting a very long time for someone to talk to."

Vite 8.2 & React 19

Three.js & Rapier 3D Physics

Groq Llama 3.3 70B & GPT-OSS 120B

Supabase Cloud Auth & Leaderboard

MIT License

🌐 Live Demo: <https://abyss-genesis-nine.vercel.app> • 🌟 GitHub: github.com/Asmit-64bit/Dark-Abyss

1. Executive Summary & Overview

Abyss is an immersive pedagogical escape room designed to test real-world computer science fundamentals through psychological tension. Players explore an atmospheric underground laboratory (*"Schrödinger's Abyss"*), inspect physics-driven terminal props, solve algorithmic decryption tasks in an integrated code execution REPL sandbox, and regulate heart-rate telemetry while surviving pursuit by an anomaly entity (*Sadako*).

The game dynamically scales challenge difficulty across standardized academic computer science domains in real time, factoring in code syntax errors, solve velocity, and hint reliance.

2. Key Innovations & Technical Features

AI Dynamic Puzzle Engine Instant inference powered by Groq (<code>openai/gpt-oss-120b</code> & <code>llama-3.3-70b-versatile</code>). Dynamically generates unique contextual escape puzzles tailored to operator performance with sub-second response times.	6-Key Failover Rotator Pool Multi-key round-robin management across 6 independent Groq keys. Automatically recovers from 429 TPM/RPM rate limits by hot-swapping active keys with zero player disruption.
Solo Solve Scoring & Telemetry +25% score multiplier awarded for unassisted independent solves. Awards precision bonuses for zero syntax errors, sub-45s speed records, and high cognitive sanity preservation ($\geq 80\%$).	Dual-Mode Cloud Leaderboard Real-time player standings powered by Supabase with automatic direct REST API fallback for static hostings (Vercel/Netlify). Ranked by Top Score, Solo Purists, and Sanity Masters.
Full Mobile & Touch Controls Dual touch controls featuring a virtual movement joystick, touch swipe camera controls, and dedicated HUD interaction triggers optimized for mobile and tablet landscape.	Psychological Sanity System Dynamic sanity gauge (0–100) tied to a procedural Web Audio heartbeat synthesizer (72–152 BPM), triggering camera grain, chromatic aberration, and subliminal apparitions.
Interactive REPL Sandbox Integrated in-game JavaScript/TypeScript execution runtime. Safely evaluates code logic with forensic feedback, syntax profiling, and error boundary recovery.	Subterranean 3D Physics Atmospheric world built on Three.js & Rapier 3D physics. Includes dynamic flashlight raycasting, pointer-lock exploration, and interactive CRT terminals.

3. Computer Science Curriculum Domains

Curriculum Domain	Technical Focus & Core Competencies
Programming Fundamentals	Lexical scoping, closure captures, type coercion, recursion depth, stack frames, core data structures.
Control Flow & Logic	Loop invariants, off-by-one boundary traps, deterministic finite automata (DFA), state machine transitions.
Web Security & APIs	SQL injection vectors, Cross-Site Scripting (XSS), JWT signature validation, REST idempotency, CORS policies.
Frontend Engineering	React 19 fiber reconciliation, useEffect lifecycle dependencies, state memoization, virtual DOM scheduling.
Networking & Cryptography	Regex token matching, CIDR subnetting, cryptographic hashing (SHA-256), Zero-Knowledge proof protocols.
Systems & Performance	Manual heap allocation (<code>malloc</code> / <code>free</code>), memory leak profiling, race conditions, mutex locks, Big-O.

4. Control Schemas

Desktop Keyboard & Mouse

Action	Input
Movement	W, A, S, D
Look / Turn	Mouse Aim (Click viewport for Pointer Lock)
Interact / Hack Terminal	E or Left Click (when targeting interactive terminal)
Toggle Flashlight	F
Jump	Spacebar
Pause / Dismiss Modal	Escape or Q
Execute Code in REPL	[RUN] button inside terminal editor

Mobile & Tablet Touch HUD

Action	Touch Gesture
Movement	Left Screen Floating Joystick
Look / Camera Aim	Right Screen Touch Swipe Surface
Interact / Terminal Access	Diegetic [INTERACT] Touch Button
Toggle Light	Diegetic [LIGHT] Touch Button
Jump Action	Diegetic [JUMP] Touch Button
Fullscreen Mode	Tap [FULLSCREEN] in Navigation Bar

5. Architecture & Technology Stack

- **Client Architecture:** Vite 8, React 19, TypeScript, Zustand global store (`gameStore` & `authStore`).
- **3D Graphics & Physics:** Three.js, React Three Fiber (R3F), Drei helper library, Rapier Physics engine.
- **AI Infrastructure:** Groq Cloud API with automated 6-key rotational failover and JSON schema enforcement.
- **Cloud Database & Storage:** Supabase PostgreSQL (custom `profiles` table, RLS security policies, and `generated_questions` database).
- **Deployment:** Continuous deployment on Vercel with zero-backend direct REST API client fallbacks.

6. Quick Start & Setup Guide

```
# 1. Clone the repository
git clone https://github.com/Asmit-64bit/Dark-Abyss.git
cd Dark-Abyss

# 2. Install dependencies
npm install

# 3. Configure environment variables (.env)
PUBLIC_SUPABASE_URL=https://your-project.supabase.co
PUBLIC_SUPABASE_ANON_KEY=eyJhbGciOi...
GROQ_MODEL=openai/gpt-oss-120b
GROQ_API_KEY=gsk_...

# 4. Run local development server
npm run dev

# 5. Build for production
npm run build
```

7. Database Architecture & Setup

- `supabase/schema.sql` : Core relational schema provisioning `profiles` and `generated_questions` tables, RLS security policies, performance indexes, and automated auth triggers.
- `npm run seed:questions` : Automated seed script populating 70+ curated academic challenges across all facility curriculum domains into Supabase.
- **Dual-Mode Leaderboard Resilience**: Real-time Supabase cloud synchronization paired with zero-latency client-side baseline NPC roster fallbacks for static and offline environments.

8. License & Acknowledgments

Licensed under the **MIT License**. Developed as a modern prototype bridging 3D interactive horror mechanics with adaptive computing education.